Alex Beaudin

a.b@eecs.berkeley.edu · (510) 990-4862 · LinkedIn · Website · Citizenship: Canada (CPT/OPT Eligible)

Education

2023 – 2028 (Expected) **UC Berkeley**, Berkeley, USA
Ph.D. Electrical Engineering and Computer Science (GPA: 4.00/4.00)
Focus: Structured Approaches to Learning and Control for Reliable Robotics
Selected Coursework: Advanced Topics in Learning and Decisions, Physics-Inspired ML, Advanced Robotic Manipulation, Stochastic Systems, Multi-Agent Systems

2019 – 2023 **McGill University**, Montreal, Canada
B.Sc. Honours Computer Science and Physics (GPA: 3.99/4.00)
Thesis: *Multi-robot Path Planning with Explicit Re-Synchronization*

Research Experience

2023 – Present **PhD Researcher**, Arcak Group, UC Berkeley
Supervisor: Prof. Murat Arcak
· *Current Research*: Exploring decision making and control under uncertainty by incorporating information dynamics in the control loop with Gaussian Processes (GPs) and Model Predictive Control (MPC).
· *Research Interests*: Adaptive and learning enabled control that leverages new data to enforce safety without sacrificing performance; optimal control and reinforcement learning (RL) for high-dimensional spaces; and applications to challenging environments, multi-agent systems, and robotics.

2022 **Undergraduate Thesis**, Mobile Robotics Lab, McGill University
Supervisor: Prof. Gregory Dudek (Director, Samsung AI Center)
***Thesis**: Multi-robot Path Planning with Explicit Re-Synchronization* — Designed and implemented multi-robot algorithms for exploration of unknown environments under communication constraints to balance efficiency and robustness without knowing the distribution of interest points a-priori.

Summer 2021 & Fall 2022 **NSERC USRA Researcher**, McGill University
Supervisor: Prof. Hsiu-Chin Lin
· *Developed* a modular simulation platform for quadrupeds to validate control algorithms, utilizing it to generate a supervised learning dataset of optimal control trajectories for training agile locomotion policies.

Publications

2025 **A. Beaudin** et al., "Action Masking for Bimanual Reinforcement Learning" (To be submitted).

2025 **A. Beaudin**, B.-A. Kristiansen et al., "Online Look-Ahead Gaussian Process Optimization for Path Planning" (Under Review).

2022 **A. Beaudin** and H.-C. Lin, "Learning Agile Paths from Optimal Control," CoRL Workshop on Agile Robotics (arXiv).

Presentations and Posters

2025 **A. Beaudin**, Y. Sonmez et al., "Diverse Experts Hurt Imitation Learning" BayLearn 2025, Poster.

Honors and Awards

2024 **Doctoral Training Scholarship + top 10 supplement** (CAD $100,000 + $6,000), Fonds de recherche du Québec

2023 **Berkeley College of Engineering PhD Fellowship** ($30,000), UC Berkeley EECS Department

2021 **Undergraduate Summer Research Award** (CAD $6,500), Natural Sciences and Engineering Research Council of Canada (NSERC)

2020 **Faculty of Science Award (Top 5%)**, McGill University

2019 **Hydro-Québec Excellence Bursary for Outstanding Students (Undergraduate Scholarship)** (CAD $15,000)

Selected Projects

Fall 2025 **Bimanual RL with Action Masking & Safety Guarantees**
This project focused on coordinated manipulation tasks requiring two arms to operate in a constrained, shared workspace. We built an RL setup in which infeasible or unsafe actions are filtered through a probabilistic model before they are passed to the policy, allowing learning to proceed without early stopping. The framework was designed in JAX with vectorized simulation to experiment with different training curricula. The goal was to study how imposing structure on the action space influences stability and sample efficiency during training, and how such structure can enable near-collision maneuvers for multiple manipulators in a shared workspace.

Fall 2025 **Benchmarking Imitation Learning and Addressing Limitations**
This project focuses on comparing different learning approaches from expert demonstrations on equal footing. We are additionally developing novel imitation learning algorithm to surpass expert demonstrations in high-dimensional settings with sparse rewards.

Fall 2025 **Criteria for Probabilistic Infinite-Time Invariance from Robust Invariance**
This project focuses on giving sufficient criteria to certify probabilistic invariance of a set. We start from a robust invariant for a bounded disturbance, and givne probabilistic guarantees of invariance for an unbounded disturbance. Involves nonlinear control theory, mathematical analysis, and stochastic systems.

2023 – Present **Multi-Step Look-Ahead GP Planner for Planning Under Uncertainty (with NTNU)**
This project explored how Gaussian Processes can be used to model environmental process during marine robot deployments, with the aim of detecting algal blooms. I developed a multi-step predictive framework that incorporates uncertainty as part of the system model, allowing the planner to reason about several sensor measurements rather than relying on single-step or overly conservative bounds. The planner uses GP-based predictions in a receding-horizon optimization fashion that remains computationally tractable for onboard deployment. The system was implemented within an existing pipeline and tested on an autonomous surface vessel used for algal bloom monitoring in Norway.

Teaching Experience

Fall 2025 **Graduate Student Instructor**, Feedback Systems (UC Berkeley)
Designed and graded labs for an upper-division undergraduate course on control theory. Implemented LQR, observers, pole placement and RL methods on various Quanser hardware setups using MATLAB and Simulink.

Fall 2024 **Graduate Student Instructor**, Linear Systems Theory (UC Berkeley)
Designed and graded assignments/exams; held office hours and recitations for graduate-level course on control theory. Prepared and delivered recitation sections on control theory fundamentals.

Fall 2021 **Undergraduate Teaching Assistant**, Honours Electricity and Magnetism (McGill University)

Professional Experience

May 2021 – Aug 2023 **Co-founder and CTO**, Layer N, Montreal, Canada
· Lead the design and implementation of a novel zero-knowledge verifiable protocol for high-throughput systems.
· Developed software ensuring correctness, verifiability, and determinism of state transitions with core components in Rust for safety and robustness. Contributed to securing investments at $50M pre-seed valuation.

Community Involvement

Ongoing **Seminar Co-organizer**, Seminar on Control, Autonomy, Learning and Embedded Systems, Berkeley EECS
Help determine and invite weekly speakers, coordinate monthly food and drinks social for attendees.

Ongoing **Lab Computer Admin**, Arcak Lab, UC Berkeley EECS
Push to obtain a lab computer for additional compute, and maintain software and access for other lab members.

Ongoing **Creator**, GGPy
An open-source Gaussain Process library in Python for ease of extensibility. Aimed at researchers using GPs.

Skills

Control & Planning: Optimal/Robust/Nonlinear/Adpative Control (MPC, LQR, SMC), Trajectory Opt. (CasADi, PyDrake), RL

Robotics & Sim: ROS/ROS2, MoveIt, Mujoco MJX, Gazebo, Docker, Git, Linux, Bash, Make

Programming: Python (PyTorch, JAX), C/C++, Rust, MATLAB

Languages: English, French (Native)

Research Areas

Robotics • Control Theory • Reinforcement Learning • Gaussian Processes • Planning Under Uncertainty • Distributed & Multi-Agent Systems • Optimization